Pi & Planck

COUNTER-INTUITIVE MATH

# Is 0.999... Equal to 1?

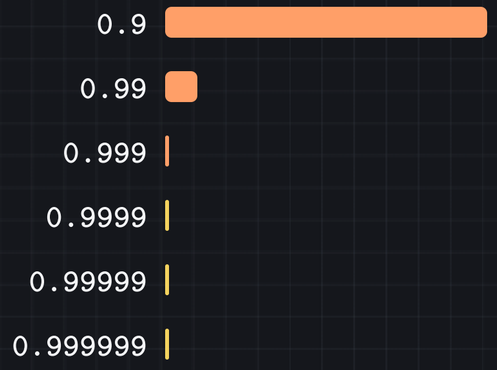
Yes, exactly. Not close to 1, not creeping toward it: the same number. Four ways to see it.

## FILLING THE BAR

One bar = 1. Add 0.9, then 0.09, then 0.009 ...




LEFT OVER




**Each step leaves a gap ten times smaller.**

## FOUR WAYS TO SEE IT

### 1 THE SERIES

$0.9 + 0.09 + 0.009 + \ldots$

$a = 0.9,\ r = 1/10$

$\text{sum} = a \div (1 - r)$

$= 0.9 \div 0.9 = 1$

MIT OpenCourseWare

### 2 THE ALGEBRA

$x = 0.999\ldots$

$10x = 9.999\ldots$

$10x - x = 9$

$9x = 9$, so $x = 1$

Subtract the two lines

### 3 THE THIRDS

$1/3 = 0.333\ldots$

$3 \times 1/3 = 1$

$3 \times 0.333\ldots = 0.999\ldots$

so $0.999\ldots = 1$

Multiply both sides by 3

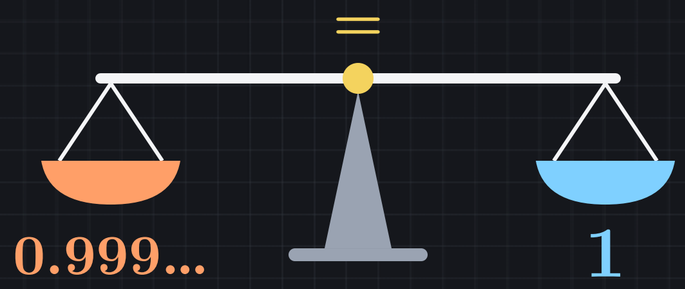
### 4 THE LIMIT

$0.9,\ 0.99,\ 0.999,\ \ldots$

get as close to 1 as you like; 1 is their least upper bound, **and that is 0.999...**

Clark University notes

## WHY IT FEELS WRONG




### Two names, one number

Like 1/2 = 0.5 = 0.4999..., some numbers have two decimal forms. 1 = 1.000... = 0.999...

### 1978

Tall and Schwarzenberger found that most students thought 0.999... was less than 1.

**1** — The sum of 0.9 + 0.09 + 0.009 + ... is 0.9 ÷ (1 − 0.1) = 1.

**$9x = 9$** — From x = 0.999... and 10x = 9.999..., subtract to get 9x = 9.

**0** — The distance from 0.999... to 1 is smaller than every positive number.

**Sources:** MIT OpenCourseWare, RES.18-001 Calculus, Chapter 10 · D. Joyce, Completeness of R, Clark University · Tall and Schwarzenberger, Mathematics Teaching 82 (1978) 44–49 · Beswick, Australian Mathematics Teacher 60(4) (2004)

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